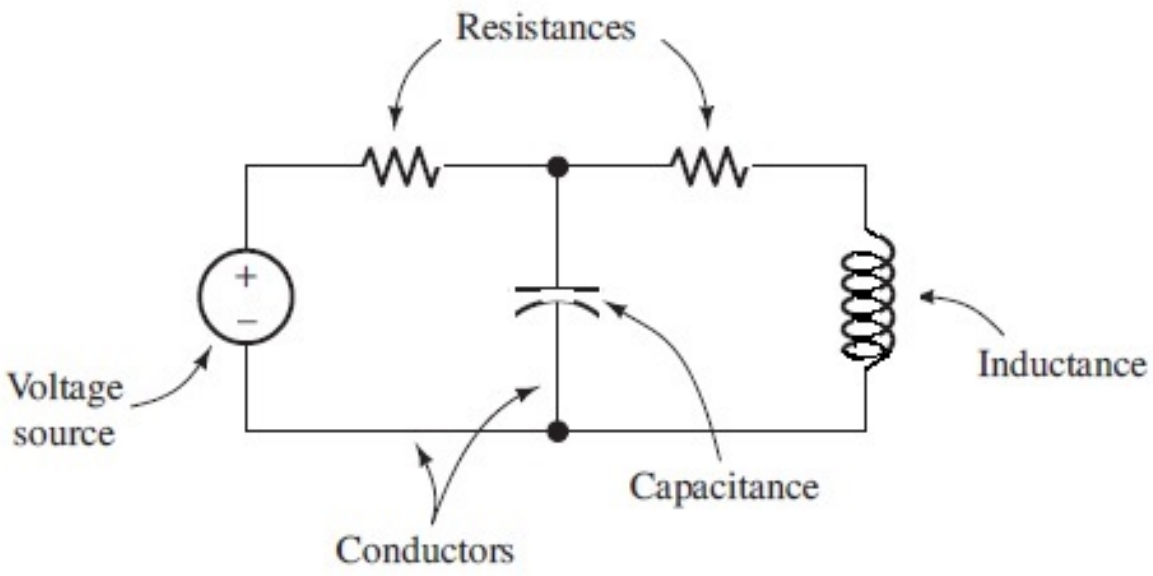
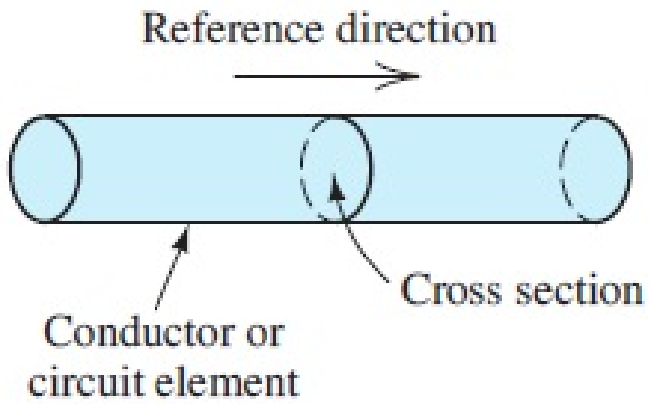
Electrical Circuits

- An electrical circuit consists of various types of circuit elements connected in closed paths by conductors. The circuit elements can be resistances, inductances, capacitances and voltage sources, among others.



Electric Current

- Electrical current is the time rate of flow of electrical charge through a conductor or circuit element. The units are amperes (A), which are equivalent to coulombs per second (C/s). The charge of an electron is $-1.602 \times 10^{-19} C$.
- Conceptually, to find the current for a given circuit element, we first select a cross section of the circuit element roughly perpendicular to the flow of current. Then, we select a reference direction along the direction of flow. Therefore, the reference direction point from one side of the corss section to the other.



Electric Current

- The current flowing through the element in the reference direction is given by

$$i(t) = \frac{dq(t)}{dt}$$

- A constant current of one ampere means that one coulomb of charge passes through the cross section each second. To find charge given current, we must integrate. Therefore, we have

$$q(t) = \int_{t_0}^{t} i(t)dt + q(t_0)$$

in which t_0 is some initial time at which the charge is known. The current that enters one end flows though the element and exists though the other end.

Example

Suppose that charge versus time for a given circuit element is given by

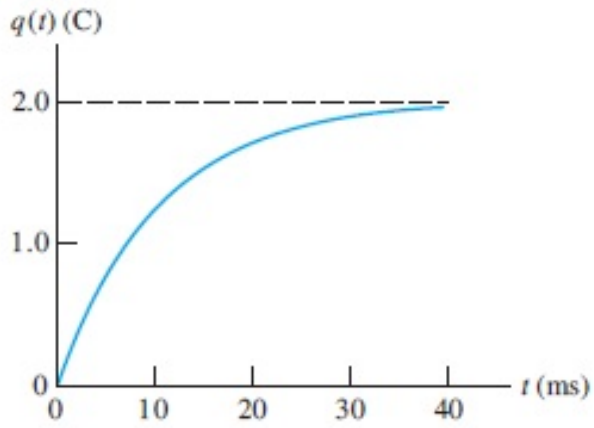
$$q(t) = \begin{cases} 0 & \text{for } t < 0 \\ 2 - 2e^{-100t}C & \text{for } t \geq 0 \end{cases}$$

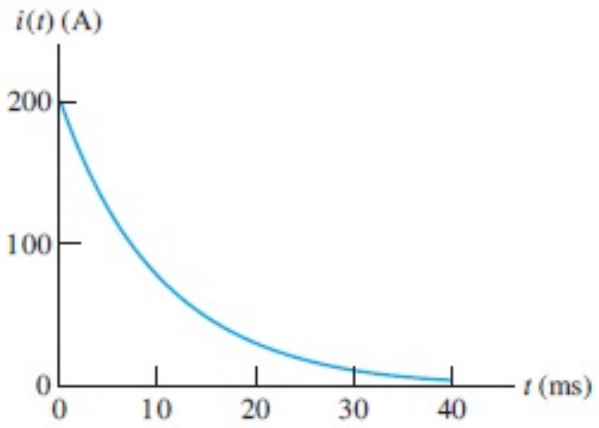
Sketch $q(t)$ and $i(t)$ versus time.

Electric Current

Solution:

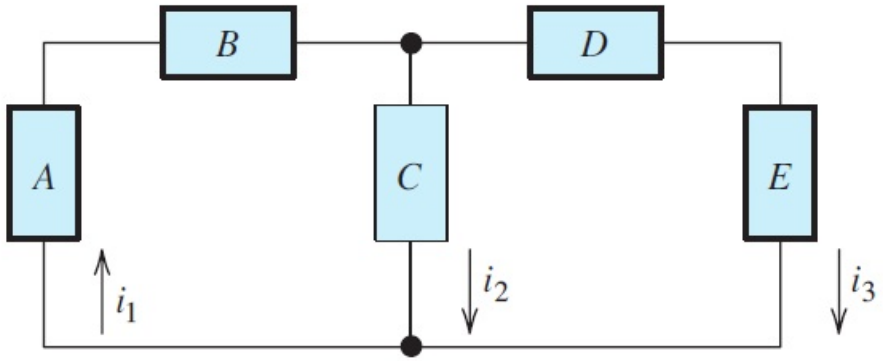
$$\begin{aligned} i(t) &= \frac{dq(t)}{dt} \\ &= 0 \quad \text{for } t < 0 \\ &= 200e^{-100t}A \quad \text{for } t \geq 0 \end{aligned}$$





Reference Directions

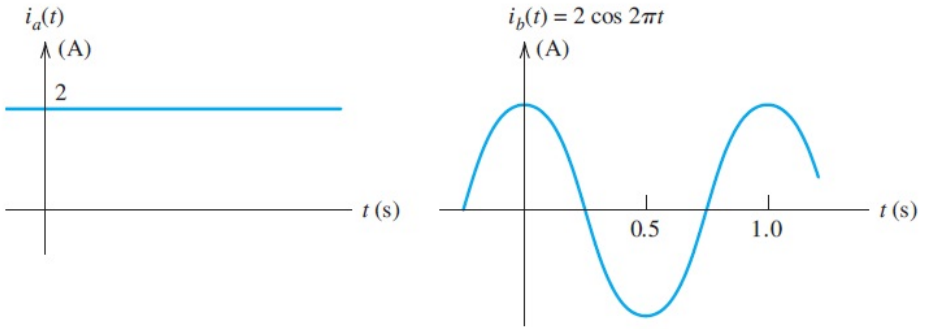
- In analyzing electrical circuits, we may not initially know the actual direction of current flow in a particular circuit element. Therefore, we start by assigning current variables and arbitrarily select a reference direction for each current of interest. It is customary to use the letter i for currents and subscript to distinguish different currents. After we solve for the current values we may find that some currents have negative values. For instance, suppose that $i_1 = -2A$ in the circuit of the following figure.



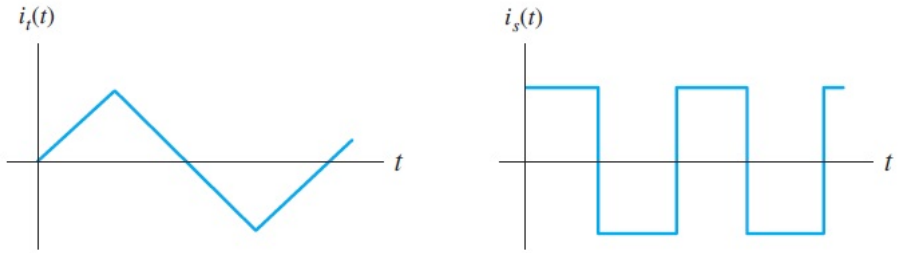
- Since $i_1 < 0$, we know that current flows in the direction opposite to the reference initially selected for i_1. Therefore, the actual current is $2A$ flowing downward through element A.

Direct Current and Alternating Current

- When a current is constant with time, we say that we have direct current, abbreviated as DC. On the other hand, a current which varies with time, reversing direction periodically, is called alternating current, abbreviated as AC.



- When $i_b(t)$ takes a negative value, the actual current direction is opposite to the reference direction for $i_b(t)$. The designation AC is used for other types of time-varying currents, such as the triangular and square waveforms.



(a) Triangular waveform (b) Square waveform

Voltages

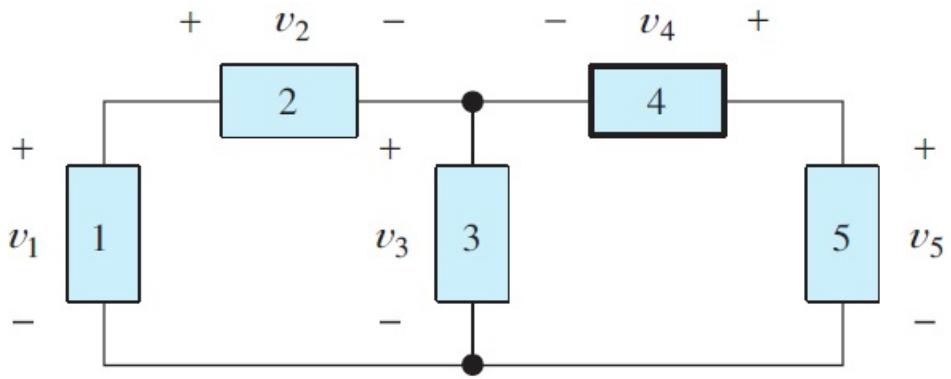
- When charge moves through circuit elements, energy can be transferred. The voltage associated with a circuit element is the energy transferred per unit of charge that flows through the element. The units of voltage are volts (V), which are equivalent to joules per coulomb (J/C).

$$V = \frac{dW}{dq}$$

where W is the energy in Joules and q is the electric charge in Coulombs.

Reference Polarities

- When we begin to analyze a circuit, we often do not know the actual polarities of some of the voltages of interest in the circuit. Then, we simply assign voltage variables choosing reference polarities arbitrarily. (Of course, the actual polarities are not arbitrary.) Next, we apply circuit principles (discussed later), obtaining equations that are solved for the voltages. If a given voltage has an actual polarity opposite to our arbitrary choice for the reference polarity, we obtain a negative value for the voltage.



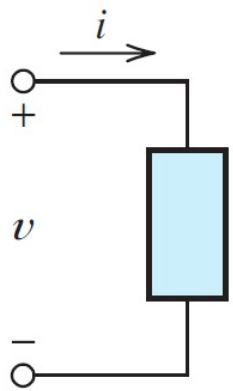
Reference Polarities

- For example, if we find that $v_3 = -5V$ in the above figure, we know that the voltage across element 3 is $5V$ in magnitude and its actual polarity is opposite to that shown in the figure.
- Voltages can be constant with time or they can vary. Constant voltages are called DC voltages. On the other hand, voltages that change in magnitude and alternate in polarity with time are said to be AC voltages. For example, $v_1(t) = 10V$ is a DC voltage whereas $v_2(t) = 10\cos(200\pi t)V$ is an AC voltage. When $v_2(t) < 0$, the actual polarity is opposite to the reference polarity.
- Double subscript notation for voltages: To represent the voltage between points a and b with the positive reference at point a, we use the notation v_{ab}.

Switches

- Switches control the currents in circuits. When an ideal switch is open, the current through it is zero and the voltage across it is determined by the remainder of the circuit. When an ideal switch is closed, the voltage across it is zero and the current through it is determined by the remainder of the circuit.

Power and Energy



- Consider the circuit element shown in the above figure. Since the current i is the rate of flow of charge and the voltage v is a measure of the energy transferred per unit of charge, the product of the current and the voltage is the rate of energy transfer. In other words, the product of current and voltage is power:

$$P = vi$$

The units of power are Joules/s or Watts.

Power and Energy

Passive Reference Configuration

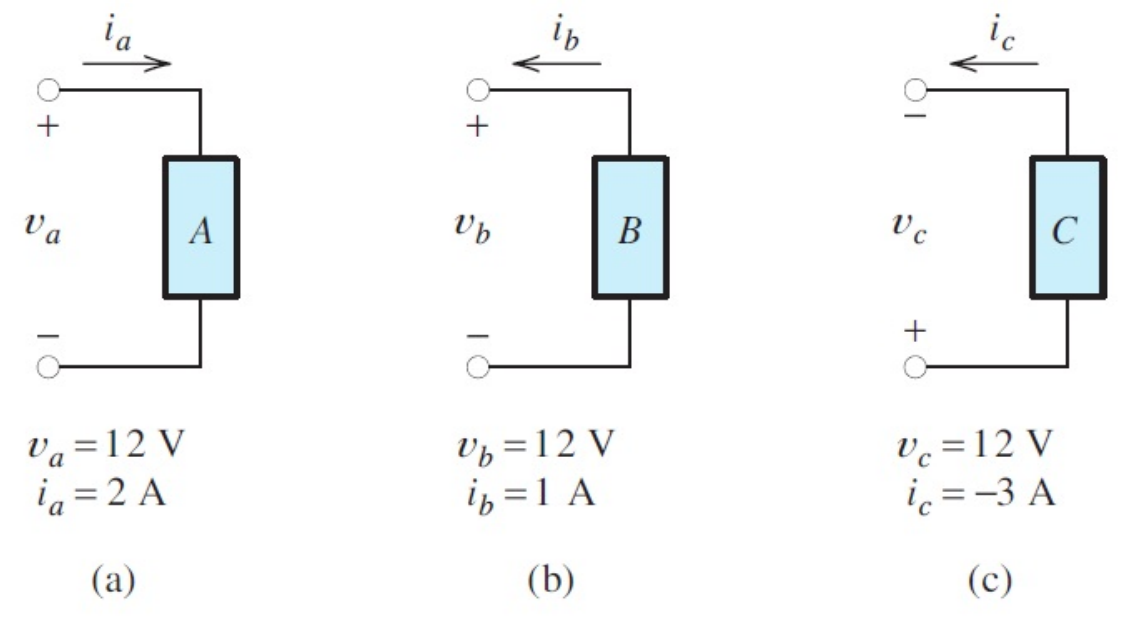
- When the current flow through an element's positive terminal, we use the equation $P = vi$. This is called passive reference configuration since a positive result for the power calculation implies that energy is being absorbed by the element. On the other hand, a negative result means that the element is supplying energy to other parts of the circuit.

Power and Energy

- If the current reference enters the negative end of the reference polarity, we compute the power as $P = -vi$. A positive calculation means the element is absorbing energy whereas a negative one means it is supplying. If the circuit element is an electrochemical battery, then positive power means that it is being charged (the battery is absorbing energy). A negative power means it is being discharged.

Power and Energy

Example



For the above figures, calculate the power for each element. If each element is a battery, is it being charged or discharged?

Power and Energy

Solution:

- In element A, the current reference enters the positive reference polarity. This is the passive reference configuration. Thus, power is computed as $p_a = v_a i_a = 12V \times 2A = 24W$ Since the power is positive, energy is absorbed by the device. If it is a battery, it is being charged.
- In element B, the current reference enters the negative reference polarity. This is opposite to the passive reference configuration. Hence, power is computed as $p_b = -v_b i_b = -(12V) \times 1A = -12W$. Since the power is negative, energy is supplied by the device. If it is a battery, it is being discharged.
- In element C, the current reference enters the positive reference polarity. This is the passive reference configuration. Thus, we compute power as $p_c = v_c i_c = 12V \times (-3A) = -36W$ Since the result is negative, energy is supplied by the element. If it is a battery, it is being discharged.

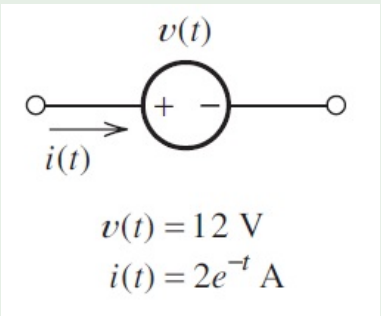
Energy Calculations

Energy Calculations

To calculate the energy w delivered to a circuit element between two time instants t_1 and t_2, we integrate power:

$$w = \int_{t_1}^{t_2} p(t)dt$$

Example



Find the expression for the power of the voltage source shown in the above figure. Compute the energy for the interval from $t_1 = 0$ to $t_2 = \infty$.

Energy Calculations

Solution:

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= 12 \times 2e^{-t} \\ &= 24e^{-t}W \end{aligned}$$

The transferred energy can be computed as

$$\begin{aligned} w &= \int_0^\infty p(t)dt \\ &= \int_0^\infty 24e^{-t}dt \\ &= [-24e^{-t}]_0^\infty = -24e^{-\infty} - (-24e^0) = 24J \end{aligned}$$

Since the energy is positive it is absorbed by the source.